

iClusterBayes Summary Report

Automated Report

Hyperparameters

```
## $n.burnin
## [1] 1000
##
## $n.draw
## [1] 1200
##
## $thin
## [1] 1
##
## $pp_cutoff
## [1] 0.5
##
## $prior_gamma
## [1] 0.1 0.1 0.1 0.1 0.1
##
## $K_values
## [1] "1 / Examined number of clusters: 2" "2 / Examined number of clusters: 3"
## [3] "3 / Examined number of clusters: 4" "4 / Examined number of clusters: 5"
## [5] "5 / Examined number of clusters: 6" "6 / Examined number of clusters: 7"
## [7] "7 / Examined number of clusters: 8" "8 / Examined number of clusters: 9"
## [9] "9 / Examined number of clusters: 10"
##
## $sdev
## [1] 0.01
##
## $beta.var.scale
## [1] 0.3
```

Summary Statistics

```
--- Summary for K = 1 ---
Z.ar (Location: fit[[1]]$Z.ar):
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.6175  0.7050  0.7158  0.7139  0.7233  0.7467

beta.ar (Location: fit[[1]]$beta.ar):
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.08667 0.13917 0.13176 0.17333 0.94417
```

```

Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1

gamma.ar (Location: fit[[1]]$gamma.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.0900 0.1417 0.1346 0.1767 0.7392
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.005000 0.008333 0.097989 0.030000 1.000000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.005000 0.008333 0.075264 0.024167 1.000000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.006667 0.015833 0.161091 0.119167 1.000000

--- Summary for K = 2 ---
Z.ar (Location: fit[[2]]$Z.ar):
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.5000 0.5475 0.5575 0.5568 0.5675 0.5983

beta.ar (Location: fit[[2]]$beta.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.0475 0.0975 0.1023 0.1408 0.9108
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1

gamma.ar (Location: fit[[2]]$gamma.ar):
  Modality 1 :

```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0500	0.1017	0.1048	0.1450	0.6933

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0008333	0.0441667	0.4049652	1.0000000	1.0000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0008333	0.0041667	0.2421158	0.3318750	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.09583	1.00000	0.68882	1.00000	1.00000

--- Summary for K = 3 ---

Z.ar (Location: fit[[3]]\$Z.ar):

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4825	0.5342	0.5575	0.5574	0.5775	0.6358

beta.ar (Location: fit[[3]]\$beta.ar):

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02417	0.06083	0.07282	0.10333	0.85583

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar (Location: fit[[3]]\$gamma.ar):

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02500	0.06333	0.07498	0.10667	0.61083

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.00000	0.02417	0.25516	0.55417	1.00000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.6942	1.0000	1.0000	0.9991	1.0000	1.0000

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.1522	0.2244	1.0000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.07917	0.51000	0.42865	0.68833	1.00000

--- Summary for K = 4 ---

Z.ar (Location: fit[[4]]\$Z.ar):

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.3908	0.4258	0.4392	0.4397	0.4525	0.5000

beta.ar (Location: fit[[4]]\$beta.ar):

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01667	0.05167	0.06417	0.09250	0.86667
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar (Location: fit[[4]]\$gamma.ar):

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01833	0.05500	0.06673	0.09833	0.56333
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.1133	0.3180	0.6533	1.0000
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.875	1.000	1.000	1.000	1.000	1.000
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.1788	0.2202	1.0000
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.3950	0.7417	0.6433	0.9908	1.0000

--- Summary for K = 5 ---

Z.ar (Location: fit[[5]]\$Z.ar):

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.8483	0.9600	0.9717	0.9690	0.9808	0.9975

beta.ar (Location: fit[[5]]\$beta.ar):

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01333	0.03583	0.06031	0.07500	0.89750
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

```

Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1

gamma.ar (Location: fit[[5]]$gamma.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.01500 0.03833 0.05982 0.07833 0.59000
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00e+00 0.00e+00 0.00e+00 4.77e-05 0.00e+00 1.00e+00
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000e+00 0.000e+00 0.000e+00 1.790e-06 0.000e+00 4.333e-02
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000000 0.0000000 0.0000000 0.0001345 0.0000000 0.1916667
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.00000 0.00000 0.00176 0.00000 1.00000

--- Summary for K = 6 ---
Z.ar (Location: fit[[6]]$Z.ar):
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
 0.4508  0.5133  0.5342  0.5333  0.5525  0.6183

beta.ar (Location: fit[[6]]$beta.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.009167 0.035000 0.048445 0.069167 0.866667
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1

gamma.ar (Location: fit[[6]]$gamma.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.01000 0.03667 0.05049 0.07333 0.52500
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.00000 0.00000 0.06576 0.03750 1.00000
  Modality 3 :

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Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.8425	1.0000	0.9028	1.0000	1.0000

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.0339	0.0000	0.8117

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.00000	0.05083	0.15137	0.26167	1.00000

--- Summary for K = 7 ---

Z.ar (Location: fit[[7]]\$Z.ar):

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.6225	0.8592	0.8842	0.8764	0.9008	0.9442

beta.ar (Location: fit[[7]]\$beta.ar):

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.006667	0.019167	0.038895	0.045000	0.847500

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar (Location: fit[[7]]\$gamma.ar):

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.006667	0.020000	0.038090	0.048333	0.523333

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.000000	0.008082	0.000000	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0000000	0.0001639	0.0000000	1.0000000

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.000000	0.002097	0.000000	1.000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.00000	0.00000	0.03163	0.00000	1.00000

--- Summary for K = 8 ---

Z.ar (Location: fit[[8]]\$Z.ar):

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.8675	0.9225	0.9383	0.9346	0.9500	0.9742

beta.ar (Location: fit[[8]]\$beta.ar):

```

Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.006667 0.016667 0.032407 0.037500 0.852500
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  1      1      1      1      1      1
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  1      1      1      1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  1      1      1      1      1      1
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  1      1      1      1      1      1

gamma.ar (Location: fit[[8]]$gamma.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.006667 0.018333 0.032206 0.040000 0.487500
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000000 0.0000000 0.0000000 0.0008431 0.0000000 1.0000000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0      0      0      0      0      0
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0      0      0      0      0      0
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.000000 0.000000 0.01489 0.000000 1.000000

--- Summary for K = 9 ---
Z.ar (Location: fit[[9]]$Z.ar):
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 0.9000  0.9642  0.9758  0.9712  0.9817  0.9942

beta.ar (Location: fit[[9]]$beta.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.005833 0.013333 0.024196 0.029167 0.878333
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  1      1      1      1      1      1

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gamma.ar (Location: fit[[9]]$gamma.ar):
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.006667 0.015000 0.025014 0.031667 0.440833
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.00e+00 0.00e+00 0.00e+00 3.56e-05 0.00e+00 1.00e+00
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000000 0.0000000 0.0000000 0.0008542 0.0000000 1.0000000
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
    0      0      0      0      0      0
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.00e+00 0.00e+00 0.00e+00 8.16e-05 0.00e+00 1.00e+00

```